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Suggest, Never Decide: Real-Time AI Assistance in Regulated Contact Centers

A governance and architecture framework for assistance that informs the agent without taking ownership of the decision.

Suggest.
Select.
Approve.

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For contact center and clinical leaders, privacy and security officers, and architects.

EXECUTIVE SUMMARY

Start from the data, not the model

Live-call assistance in healthcare is a PHI system first and an AI system second. Establish its data boundaries before choosing its model.

The assistant suggests, the agent selects, the system of record approves.

Data minimization

Prevent unnecessary copies by construction.

Reasons, not scores

Give agents something they can check.

Measurement as governance

Separate what was shown from what was kept.

Pre-registered evaluation

Commit to the bar before measuring.

Clinical gating

Require named owners and evidence.

Engineering controls

Fail before a call does.

This paper is design guidance, not legal advice. Obtain organization-specific privacy, security and legal review.

LIVE-CALL AI IS A PHI SYSTEM FIRST

A transcript of a member describing their health is PHI the moment it exists.

The question is how to stop a dozen copies from existing.

Make minimum necessary operational

Express HIPAA's minimum-necessary principle as a per-component inventory. Record what enters each component, what it holds, how long it remains and who is accountable.

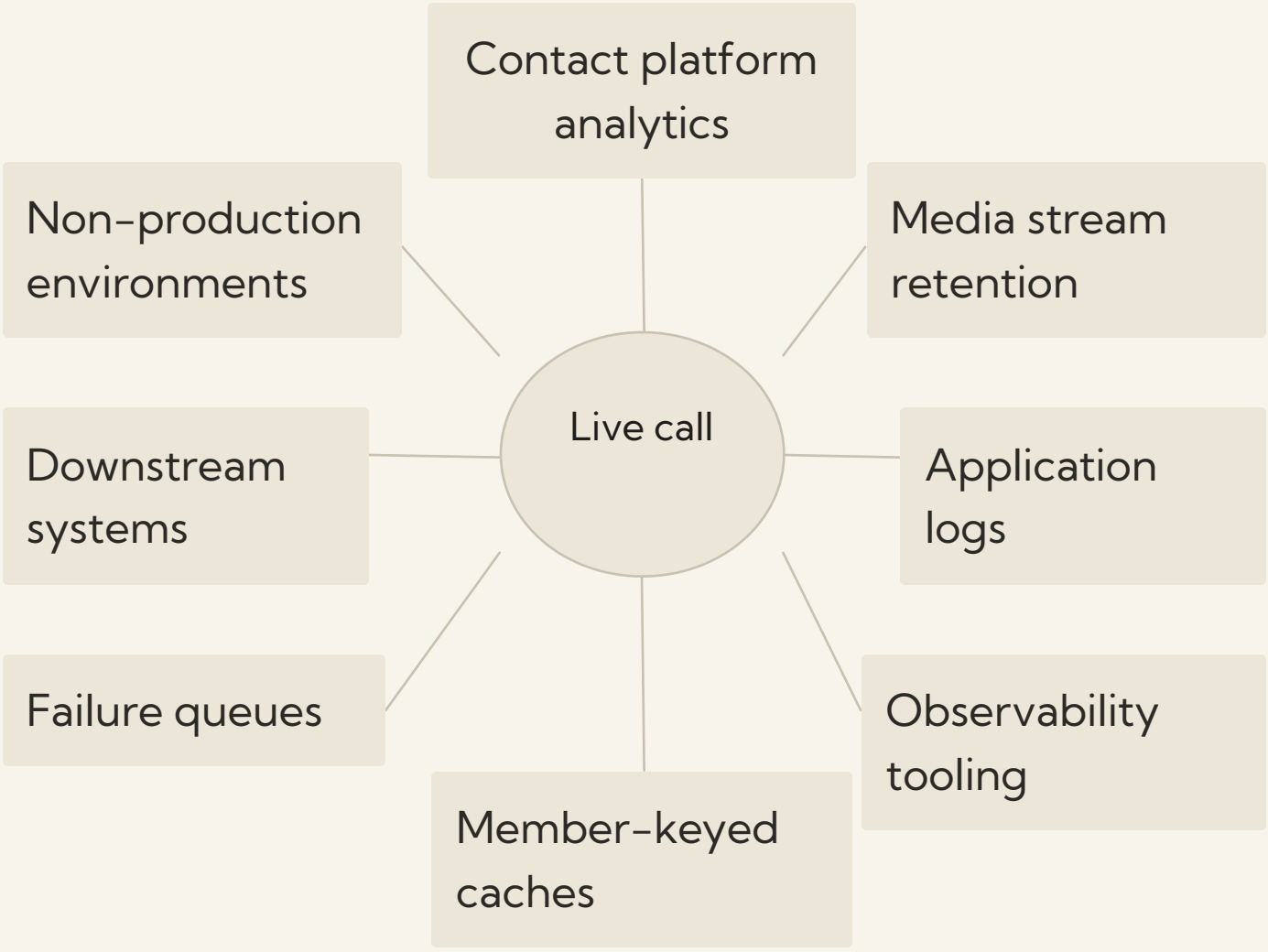
Include infrastructure that observes or retries the call path. A short-lived transcript still needs controls while it exists; avoiding durable storage does not remove that obligation.

Reference: 45 CFR 164.502(b) and 164.514(d). Applicability and exceptions require legal review.

COPY INVENTORY

Where unplanned copies of a call come from

Inspect the surrounding services, not only the assist application. Defaults can create retention that the architecture never intended.



Every destination needs an explicit purpose, retention rule and owner. "Temporary" is not a retention policy.

ARCHITECTURE SEQUENCE

Why data decisions come before model decisions

The model is the easiest part to change later; data flows are the hardest.

1

Decide on day one

Set the rule that the assist service persists nothing. Deployment, logging, retry handling and observability must then follow that boundary.

2

Defer until day ninety

The same decision becomes a search for caches, logs and queues nobody listed. Removing a database does not remove copies already created elsewhere.

3

Approve the footprint before the model

Use the inventory to review each integration and vendor boundary. Model substitution should fit the approved data contract rather than silently expand it.

The footprint is a design constraint, not a cleanup task after a successful pilot.

RESPONSIBILITY BOUNDARY

The design rule: suggest, select, approve

Keep relevance, judgment and authorization with different owners. A useful suggestion is not permission to share it.

Recommend	Select	Approve
Assist service	Agent	System of record
Owns relevance. Never owns the decision to share.	Owns judgment in the seconds a call allows.	Applies consent and channel rules, then sends.

The system of record holds the durable communication record. Selecting an item in the assist panel must not bypass that system's checks.

CLINICAL SCOPE

Scope set by clinical owners, in three phases

Each phase changes what the service may know or do. Progression requires a new scope decision, not simply a more capable model.

01 Conversation only

Suggest approved library documents from the conversation. No member record access.

02 Conversation plus member context

Context is pushed at call start and dropped at hang-up. Define the permitted fields and eligibility criteria before enabling access.

03 Care recommendations

LOCKED PENDING CLINICAL APPROVAL

Deliberately undesigned until clinical owners specify, evaluate and sign off the use case.

A technical integration does not confer clinical authorization.

REFERENCE ARCHITECTURE

Six separately owned parts

Separate the live processing path from durable decision records. Ownership follows those boundaries.

CALL-SCOPED ZONE · RELEASE AT HANG-UP

Speech

Audio → segments

Business logic

Session → suggestions

Models

Inference only

Knowledge index → business logic

Approved library content and embeddings support retrieval. The index is not a destination for call transcripts.

System of record

Agent selection → consent and channel checks → communication record.

Measurement

Shown record + kept record → evaluation without transcript.

Speech, session and model processing use call content only within the approved live-call window. Durable destinations receive decision records, never the transcript.

DATA FOOTPRINT

Different data, different time

What each part holds, for how long, and who owns it: the most useful single artifact a program can produce.

Part	What it holds	How long	Owner
Speech	Audio and transcript segments in flight	Call duration plus shortest restart retention	Contact platform team
Business logic	Session state, suggestions shown, agent actions	Call duration, in memory only	Assist service team
Models	Nothing	None	Data science team
Knowledge index	Approved library content and embeddings	Until content is retired	Clinical and content owners
System of record	Selections kept, items sent, communications	Existing record retention	CRM or care platform owner
Measurement	What was shown and kept, closed-set codes; no transcript	Set by evaluation program	Measurement owner with privacy review

"Nothing" and "None" describe model retention, not the absence of transient inference processing. Validate provider behavior against this contract.

MINIMIZATION

Data minimization by construction

The assist service should have no durable place to put call content. Apply that constraint across its runtime and supporting tooling.

No persistence surface

No volume, database, cache or object storage, anywhere in the assist service's call-content path.

No diagnostic copy

No log line carries call content. Payload capture is off across the path, including observability and failure handling.

No hidden member store

No member-keyed lookup. Non-production is treated as production for these controls.

Two payloads, separate destinations

Transcript → transient processing only.

Decision record → approved durable systems.

Keep the decision record minimal and governed. "No transcript" does not automatically mean "no PHI."

PLATFORM BOUNDARY

Controls at the contact platform edge



Keep one transcription path

Turn duplicate transcription off. Set stream retention to the minimum needed for a restart.



Reject unknown start fields

Enforce the start message contract, keeping the caller number out.



End without identifiers

Detect hang-up without returning caller identifiers.



Constrain event handling

No dead-letter destination on event handlers. Use least-privilege workload identity.



Review audio crossings early

Audio crossing between cloud providers needs its own security approval.

AGENT EXPERIENCE

Reasons, not confidence scores

Scores invite automation bias, imply unearned calibration and give the agent nothing to act on. Show why an item is listed, in language checkable in a second or two.

ILLUSTRATIVE ASSIST PANEL

Approved library document

Reason shown: matches the topic discussed in this call.

Agent reviews the document before selecting it.

LIBRARY'S CLOSEST GUESS

Near-match library document

Reason shown: related topic; an exact library match was not found.

Near-match text is not stored in the record.

Make verification easier than deference

Presenting information rather than a recommendation is linked to less over-reliance in the automation-bias literature. Keep evidence visible and leave judgment with the agent.

Reference: Goddard et al., JAMIA, 2012. Panel text is illustrative, not a clinical recommendation.

MEASUREMENT DESIGN

Measurement as governance: shown versus kept

Two independently written halves make the evaluation record harder to reshape after the fact. Neither system can rewrite the other's half.

ASSIST SERVICE WRITES

Shown

Which suggestions were displayed.

SYSTEM OF RECORD WRITES

Kept

Which selections were retained.

Join the halves → measurement store

Rejections are the difference between shown and kept, reconciled within the same evaluation context.

Six closed-set decline codes

Agents choose from an approved set with no free text. Free text about a member is PHI. Export the governed decision fields, never the transcript.

INTERPRETATION

Three failures a naive acceptance rate hides

Each case needs a different owner to act. A single acceptance rate mixes exposure, system health and retrieval quality.

Case	What it means	Who acts
Panel never viewed	Agent did not look; suggestion quality unknown	Operations and training
Degraded panel	Transcription lag, model timeout or index problem	Engineering
Document sent but never suggested	A recall gap; the most valuable signal	Content owners and data science

Preserve the distinction in reporting

Capture a viewed signal, stamp outages and reconcile sent documents with suggestions. Without those signals, apparent rejection can be misread as poor relevance.

EVALUATION CONTRACT

Pre-registered evaluation and clinical gating

Every threshold goes in a decision register with its adoption condition before it is measured. Latency belongs beside relevance.

1

Predict

Name the setting, data and expected behavior.

2

Commit

Record the metric, bar, adoption condition and decision owner.

3

Measure

Evaluate against the registered conditions, including latency.

4

Sign-off

Name who approves, on what evidence, and by when.

5

Enable

Clinical features remain wired but off in production until approval.

Post-go-live review loops back to the decision register when evidence, scope or operating conditions change.

ENGINEERING GATES

Engineering controls and hosted-model pilots

Refuse a bad use case at startup, before a live call depends on it.

1 Validate → fingerprint

Import-time validation and schema fingerprints check the use case contract.

2 Readiness → canary

Readiness gates and a canary vector verify that dependencies behave as expected.

3 Any failed gate → Refused

Do not admit the use case to the call path.

Constrain hosted-model pilots

- Tenants barred from hosted models run with the model absent, not just disabled.
- Require token authentication before tester access. Pin revision IDs on suggestions.
- Show generative extraction beside evidence, with confirm or flag. Keep it off by default pending BAA and privacy review; measure before widening.

A pilot remains subject to the same data boundaries as production.

IMPLEMENTATION

Implementation checklist and open questions

Before build

- Phase scope
- Footprint table
- Audio crossing review
- Record format
- Decline codes
- Decision register

During build

- No storage or content in logs
- Payload capture off
- Edge controls
- Canary
- Revision IDs
- Outage stamping

Before go-live

- Authentication
- Non-production parity
- Clinical sign-off
- Approved trigger phrases
- Measurement separating the three cases

Open early

Who owns clinical sign-off? Where is the audio boundary? What proves the panel was viewed? What is the latency budget? Which member-context criteria apply? Who will define care recommendations?

About the author

Aejaz Sheriff Quaraishi is the author of this white paper on governance and architecture for real-time assistance in regulated contact centers.

Nothing in this paper is legal advice. Validate requirements with your legal, privacy, security and clinical owners.

References

Goddard et al. (2012), "Automation bias: a systematic review of frequency, effect mediators, and mitigators," JAMIA.
HIPAA: 45 CFR 164.502(b), 164.514(d), 164.502(e), 164.504(e).
NIST AI RMF 1.0; NIST AI 600-1, Generative Artificial Intelligence Profile.